Government College of Engineering, Jalgaon (An Autonomous Institute of Government of Maharashtra)	
Name :	**PRN :**
Class : L. Y. B.Tech Computer	**Batch:**
Subject : CO456U Cloud Computing Lab	**Sem :** VIIIth
Course Teacher: Prof. S. D. Cheke	**Academic Year :** 2025-26
Date of Performance :	**Date of Completion :**

Practical No. 7

Aim: Create and deploy lambda function that takes request and put that data to dynamodb.

Theory :

Objective: To create an AWS Lambda function that receives a request and stores the data into a DynamoDB table.

What is AWS Lambda?

AWS Lambda is a serverless computing service that allows you to run code without managing servers.

- Executes code on demand
- Automatically scales
- Charges only for execution time

What is DynamoDB?

Amazon DynamoDB is a fully managed NoSQL database service by AWS.

- Stores data in key-value format
- Highly scalable and fast
- No server management required

What is Serverless Architecture?

Serverless Architecture is a model where the cloud provider manages infrastructure, and developers focus only on code.

How It Works

1. User sends request (via API Gateway or test event)
2. Lambda function processes request
3. Data is inserted into DynamoDB table

Requirements

- AWS Account
- IAM Role with DynamoDB access
- Basic knowledge of Python

Procedure / Steps

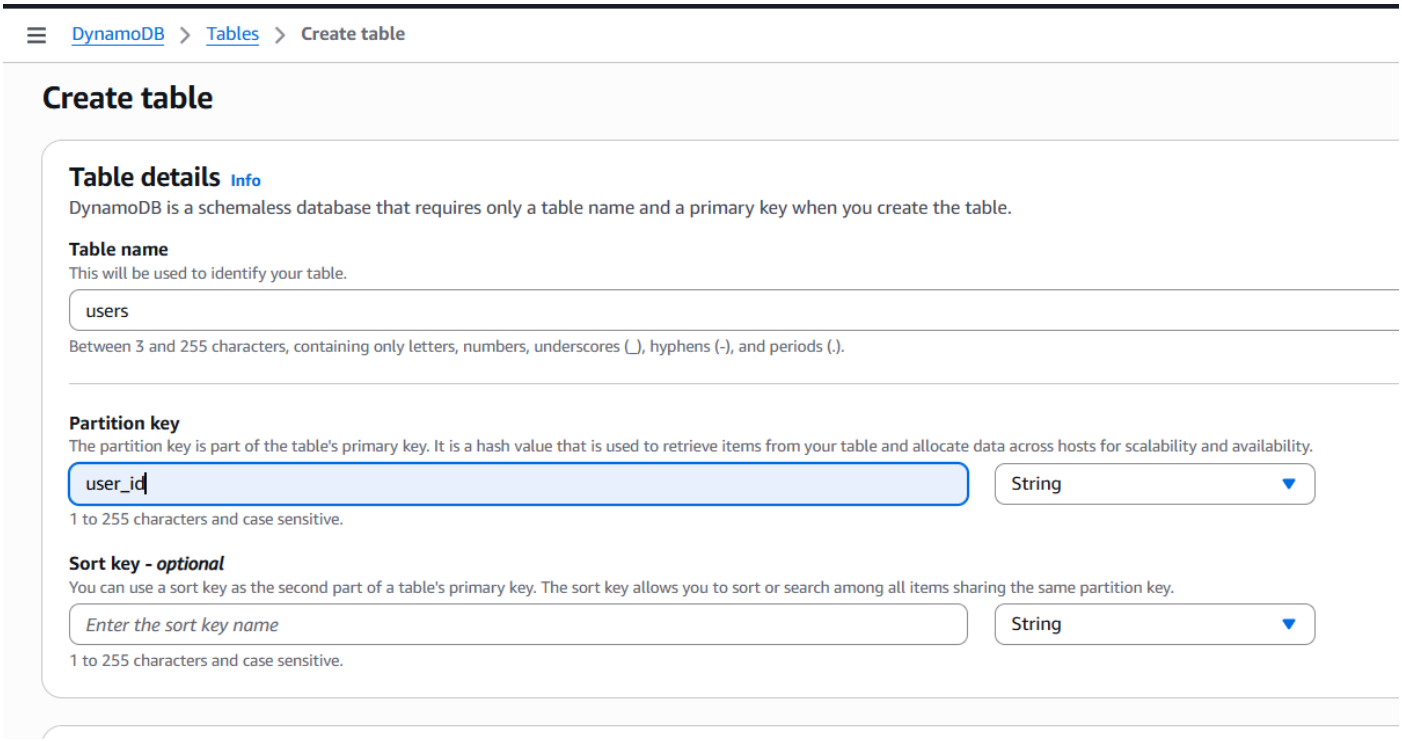
Step 1: Open DynamoDB

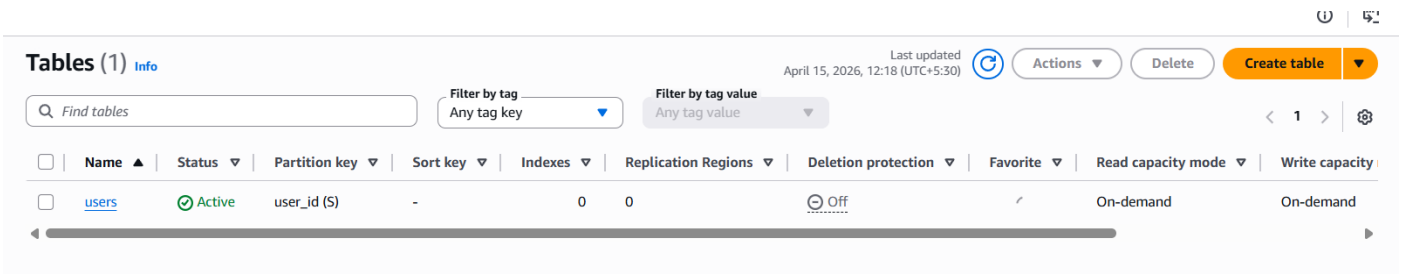
1. Go to AWS Console
2. Search for DynamoDB
3. Click **Create Table**

Step 2: Configure Table

- **Table Name**: users
- **Primary Key**: user_id (String)

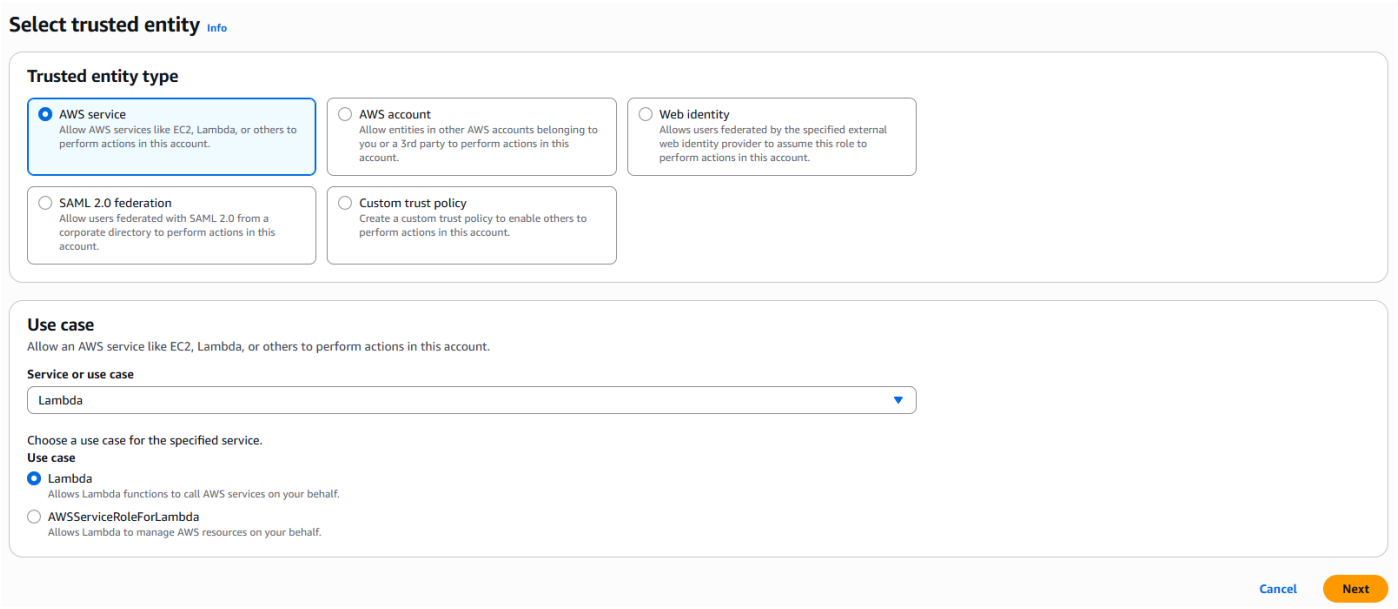
Click **Create Table**



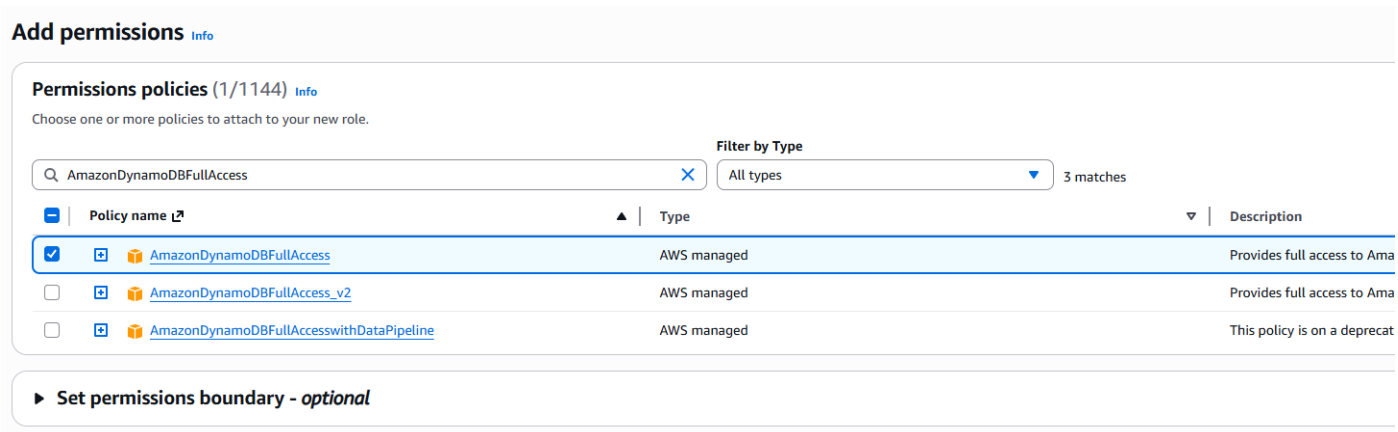


Step 2: Create IAM Role

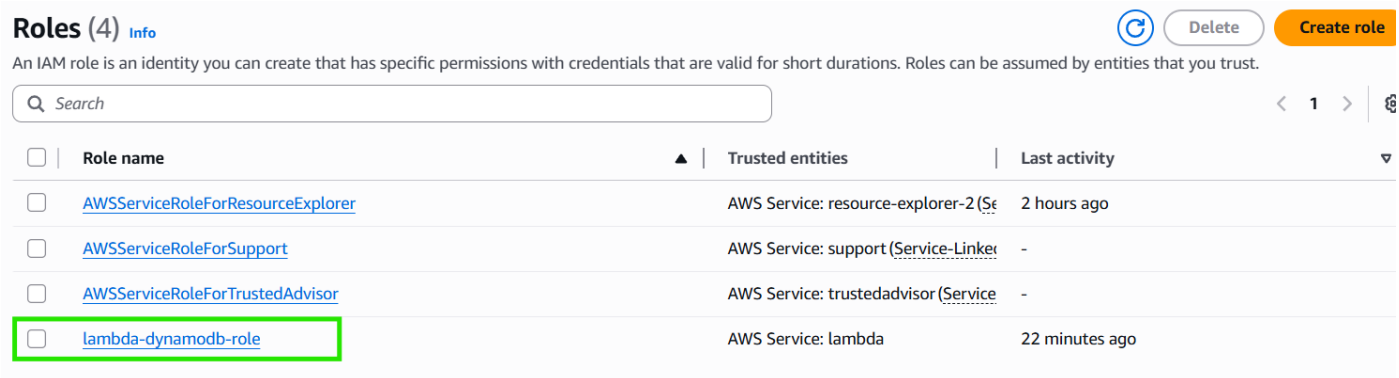
1. Go to IAM → Roles
2. Click **Create Role**
3. Select:
    1. Trusted entity: **AWS Service**
    2. Use case: **Lambda**



4. Attach policy:
    1. **AmazonDynamoDBFullAccess**



5. Name: lambda-dynamodb-role
6. Click **Create Role**

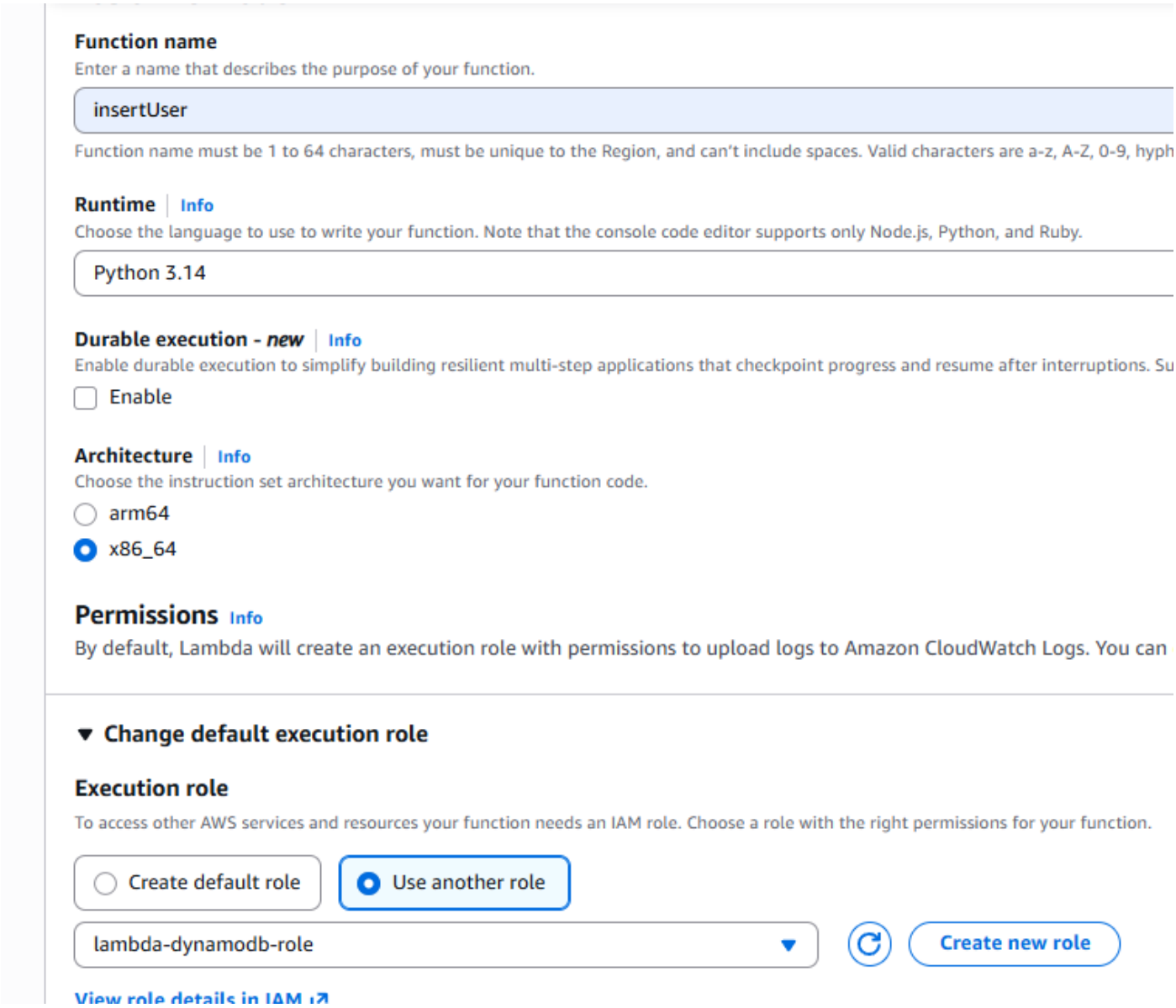


Step 3: Open Lambda

1. Go to AWS Lambda
2. Click **Create Function**

Step 4: Configure Function

- **Function Name**: insertUser
- **Runtime**: Python 3.x
- **Execution Role**: Create new role with DynamoDB full access



Write Lambda Code

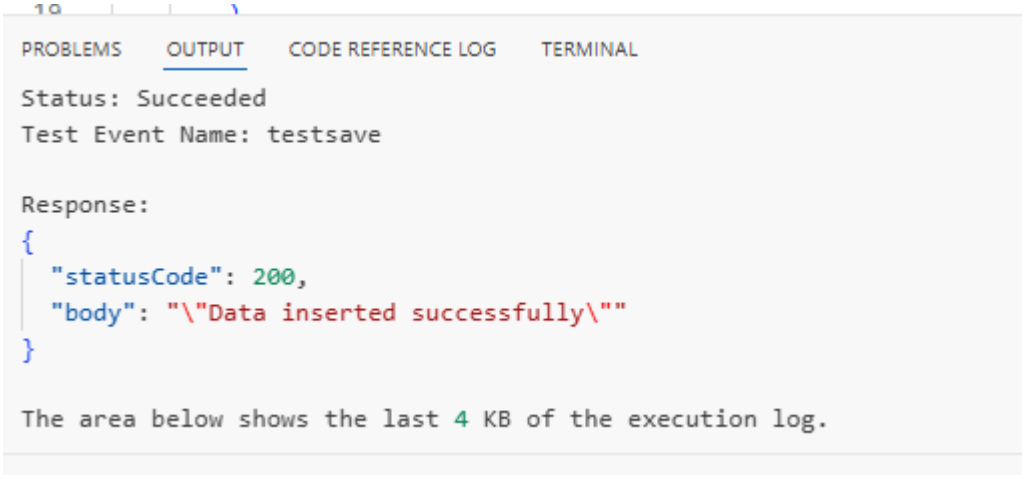
```
import json

import boto3

dynamodb = boto3.resource('dynamodb')

table = dynamodb.Table('Users')

def lambda_handler(event, context):

    try:

        # Get data from request
```

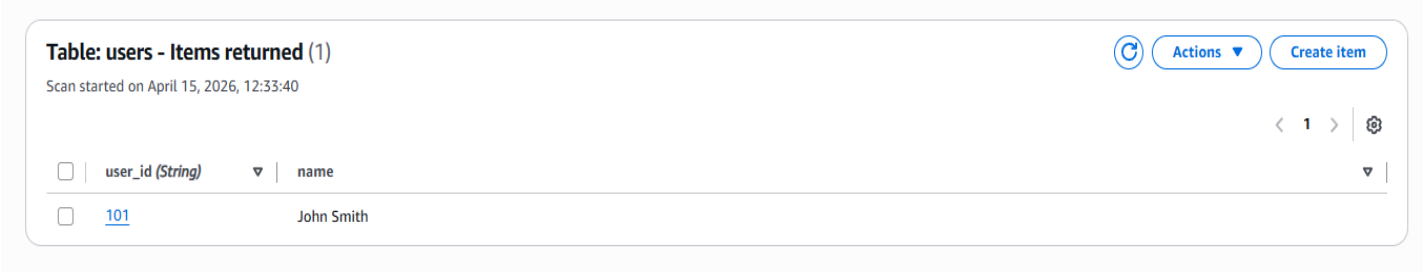
```
        user_id = event['user_id']
        name = event['name']
        # Insert into DynamoDB
        table.put_item(
            Item={
                'user_id': user_id,
                'name': name
            }
        )
        return {
            'statusCode': 200,
            'body': json.dumps('Data inserted successfully')
        }
    except Exception as e:
        return {
            'statusCode': 500,
            'body': str(e)
        }
```

Create Test Event

1. Click **Test**
2. Configure test event:

```
{
  "user_id": "101",
  "name": "John Smith"
}
```





Conclusion:

In this way, we have successfully implemented and completed the creation and deployment of a Lambda function that stores request data into DynamoDB.

Prof. S. D. Cheke
Course Teacher